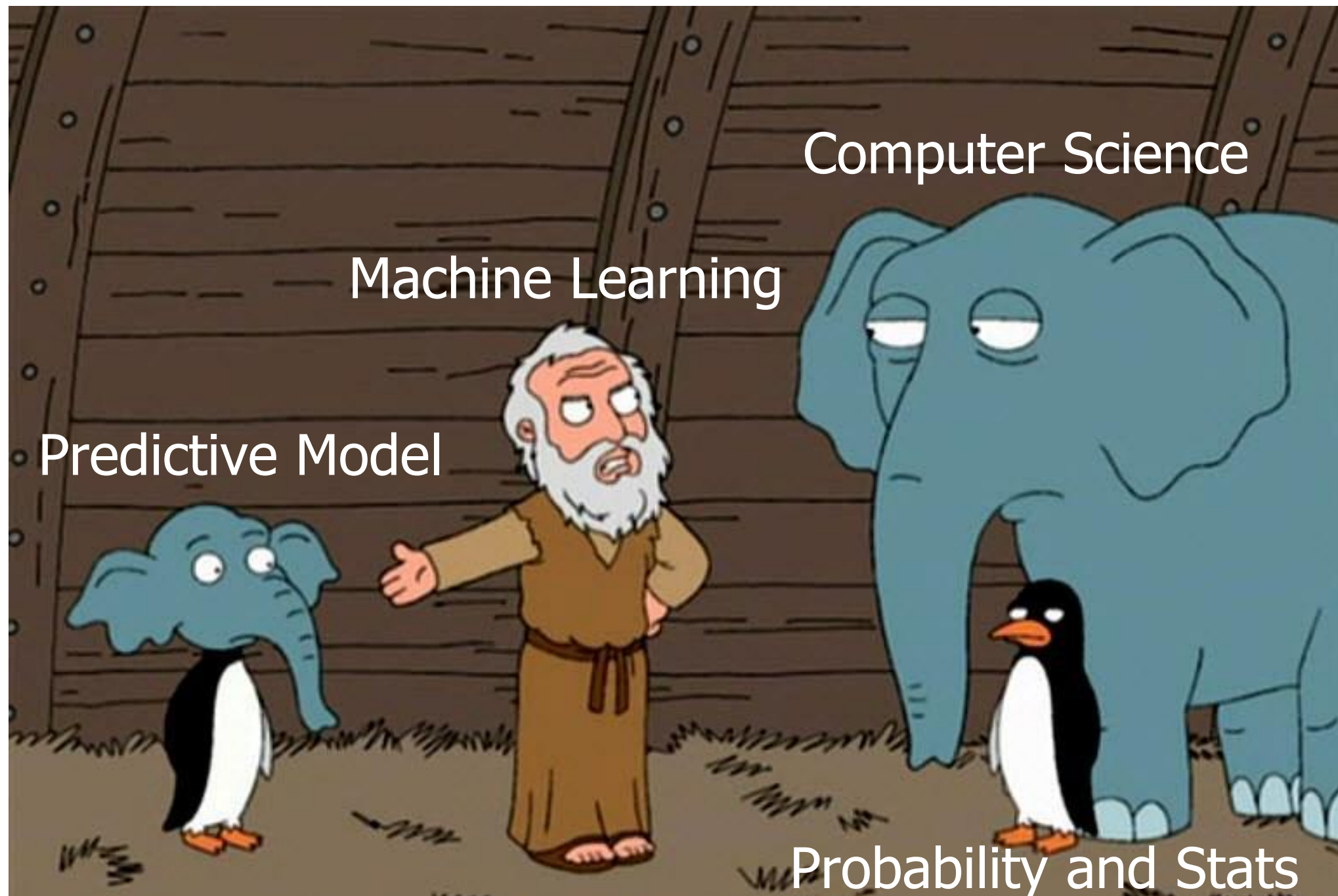


Probability and Statistics

Mahdi Roozbahani
Georgia Tech



Computer Science

Machine Learning

Predictive Model

Probability and Stats

Outline

- Probability Distributions 
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation

Probability

- A **sample space S** is the set of all possible outcomes of a conceptual or physical, repeatable experiment. (S can be finite or infinite.)
 - E.g., S may be the set of all possible outcomes of a dice roll: S
(1 2 3 4 5 6)
 - E.g., S may be the set of all possible nucleotides of a DNA site: S
(A C G T)
- E.g., S may be the set of all possible time-space positions of an aircraft on a radar screen.
- An **Event A** is any subset of S
 - Seeing "1" or "6" in a dice roll; observing a "G" at a site; UA007 in space-time interval



Three Key Ingredients in Probability Theory

A **sample space** is a collection of all possible **outcomes**

Random variables X represents **outcomes** in sample space

Probability of a random variable to happen $p(x) = p(X = x)$

$$p(x) \geq 0$$

$$0 \leq p(x) \leq 1$$

Continuous variable

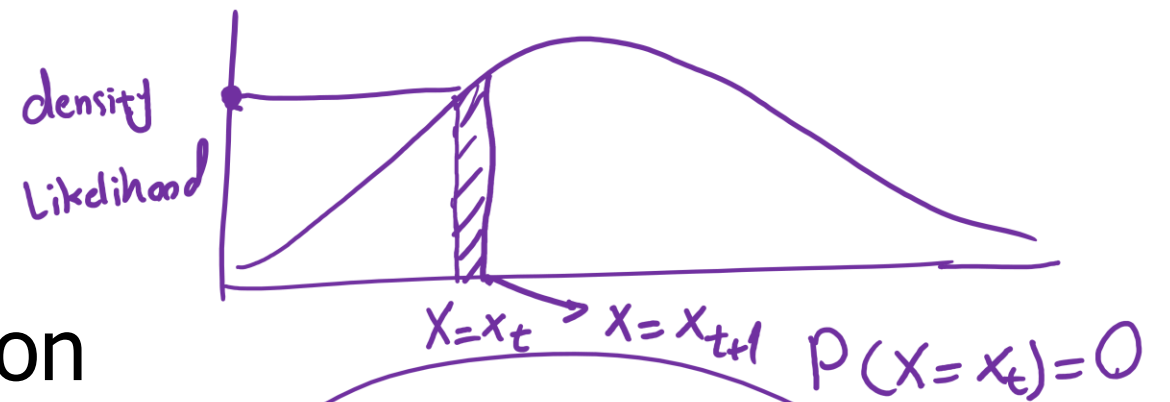
Continuous probability distribution

Probability density function pdf

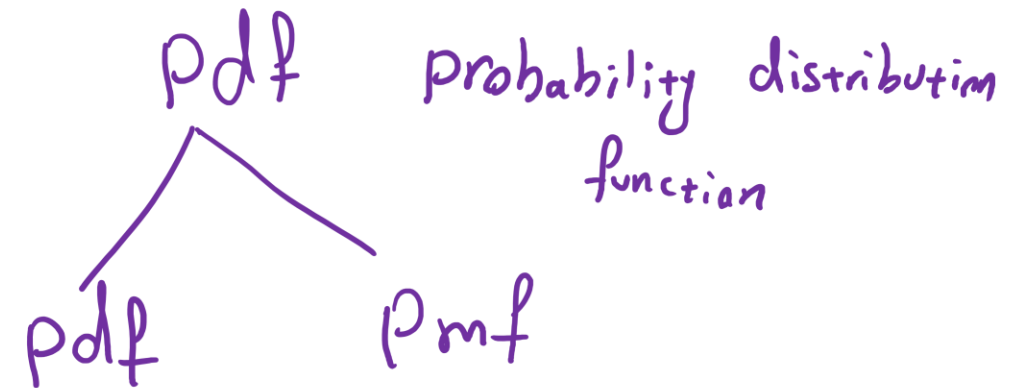
Density or likelihood value

Temperature (real number)

Gaussian Distribution



$$\int_x p(x) dx = 1$$



Discrete variable

Discrete probability distribution

Probability mass function pmf

Probability value

Coin flip (integer)

Bernoulli distribution

$$\sum_{x \in A} p(x) = 1$$

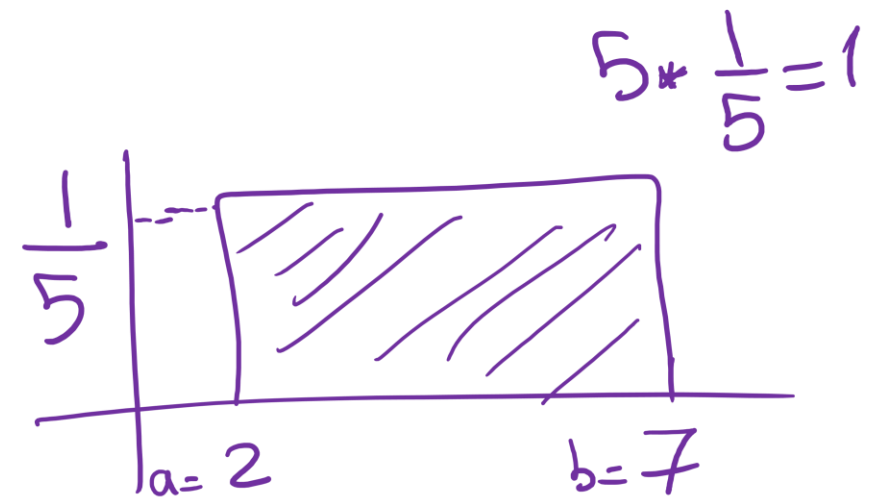
Continuous Probability Functions

• Examples: $\Theta \rightarrow$ Parameters

$$\{b, a\} \in \Theta$$

• Uniform Density Function:

$$f_x(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



• Exponential Density Function: $\{\mu\} \in \Theta$

$$f_x(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}} \quad \text{for } x \geq 0$$

$$F_x(x) = 1 - e^{-\frac{x}{\mu}} \quad \text{for } x \geq 0$$

μ { Arithmetic avg
weighted avg or expected value

• Gaussian(Normal) Density Function $\{\mu, \sigma\} \in \Theta$

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\mu = \frac{\sum_{i=1}^N x^{(i)}}{N}$$

$$\sigma^2 = \frac{1}{N} \sum (x^{(i)} - \mu)^2$$

Discrete Probability Functions

- Examples:

- Bernoulli Distribution:

- $$\begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \end{cases}$$

In Bernoulli, just a **single** trial is conducted

- Binomial Distribution:

- $$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

k is number of successes

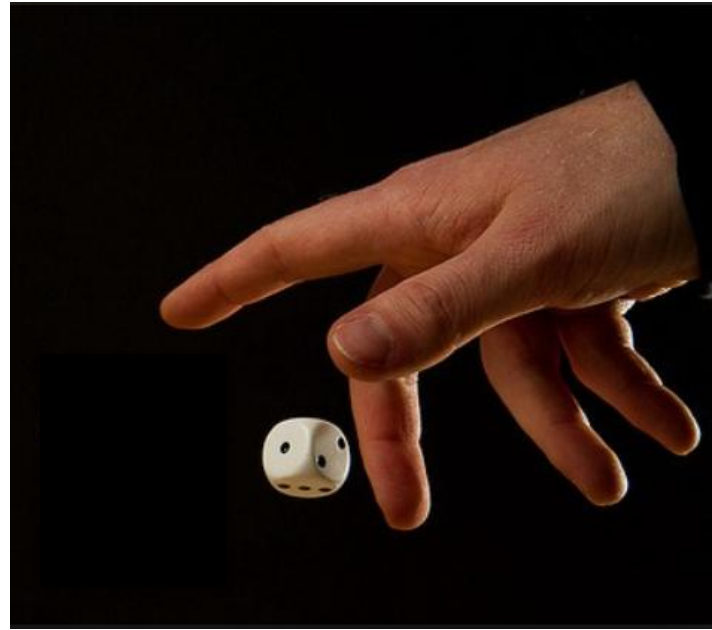
n-k is number of failures

$\binom{n}{k}$ The total number of ways of selection **k** distinct combinations of **n** trials, **irrespective of order**.

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Example



X = Throw a
dice



Y = Flip a coin

$\mathbf{X}$ and $\mathbf{Y}$ are random variables

$\mathbf{N}$ = total number of trials

n_{ij} = Number of occurrence

		$\mathbf{X}$						
		$x_{i=1} = 1$	$x_{i=2} = 2$	$x_{i=3} = 3$	$x_{i=4} = 4$	$x_{i=5} = 5$	$x_{i=6} = 6$	C_j
$\mathbf{Y}$	$y_{j=2} = tail$	$n_{ij} = 3$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 5$	$n_{ij} = 1$	$n_{ij} = 5$	20
	$y_{j=1} = head$	$n_{ij} = 2$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 1$	15
	C_i	5	6	6	7	5	6	N=35

X**C_j**
 $x_{i=1} = 1$ $x_{i=2} = 2$ $x_{i=3} = 3$ $x_{i=4} = 4$ $x_{i=5} = 5$ $x_{i=6} = 6$
Y $y_{j=2} = tail$
 $y_{j=1} = head$
 C_i

$n_{ij} = 3$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 5$	$n_{ij} = 1$	$n_{ij} = 5$	20
$n_{ij} = 2$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 2$	$n_{ij} = 4$	$n_{ij} = 1$	15
5	6	6	7	5	6	N=35

$$P(Y=t, X=4) = \frac{5}{35} = \frac{n_{ij}}{N}$$

$$P(Y=t) = \frac{20}{35} = \frac{C_j}{N}$$

$$P(X=4) = \frac{7}{35} = \frac{C_i}{N}$$

$$P(Y=t | X=4) = \frac{5}{7} = \frac{n_{ij}}{C_i}$$

$$P(X=4 | Y=t) = \frac{5}{20} = \frac{n_{ij}}{C_j}$$

$$P(A, B, C) = P(A | B, C) P(B, C)$$

$$P(Y, X) = \frac{n_{ij}}{N} = \frac{n_{ij}}{C_j} \frac{C_j}{N} = \frac{n_{ij}}{C_i} \frac{C_i}{N} = P(A, B | C) P(C)$$

$$P(Y, X) = P(X | Y) P(Y) = P(Y | X) P(X)$$

Probability:

$$p(X = x_i) = \frac{c_i}{N}$$

Joint probability:

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Conditional probability: $p(A|B) = \frac{p(A, B)}{p(B)}$

$$p(Y = y_j | X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)} = \frac{n_{ij}}{c_i}$$
$$p(A, B, C) = p(A|B, C) p(B, C) = p(A, B|C) p(C)$$

Sum rule

$$p(X = x_i) = \sum_{j=1}^L p(X = x_i, Y = y_j) \Rightarrow p(X) = \sum_Y P(X, Y)$$

Product rule

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i} \frac{c_i}{N} = p(Y = y_j | X = x_i) p(X = x_i)$$

$$p(X, Y) = p(Y|X)p(X)$$

Conditional Independence

- Examples: $P(h, f, v, d) = P(h | f, v, d) \cdot P(f, v, d) =$
 $P(h | f, d) \cdot P(f | v, d) \cdot P(v, d)$
 $P(h | f, d) \cdot P(f | v) \cdot P(v | d) \cdot P(d)$
 $P(h | f, d) \cdot P(f | v) \cdot P(v) \cdot P(d)$
- $P(\text{Virus} | \text{DrinkBeer}) = P(\text{Virus})$
 iff **Virus** is independent of **Drink Beer**
- $P(\text{Flu} | \text{Virus}, \text{DrinkBeer}) = P(\text{Flu} | \text{Virus})$
 iff **Flu** is independent of **Drink Beer**, given **Virus**

$$P(\text{Headache} | \text{Flu}, \text{Virus}, \text{DrinkBeer}) \\ = P(\text{Headache} | \text{Flu}, \text{DrinkBeer})$$

iff **Headache** is independent of **Virus**, given **Flu** and **Drink Beer**

Assume the above independence, we obtain:

$$P(\text{Headache}, \text{Flu}, \text{Virus}, \text{DrinkBeer}) \\ = P(\text{Headache} | \text{Flu}, \text{Virus}, \text{DrinkBeer}) P(\text{Flu} | \text{Virus}, \text{DrinkBeer}) \\ P(\text{Virus} | \text{DrinkBeer}) P(\text{DrinkBeer}) \\ = P(\text{Headache} | \text{Flu}, \text{DrinkBeer}) P(\text{Flu} | \text{Virus}) P(\text{Virus}) P(\text{DrinkBeer})$$

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- Bayes' Rule 
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Bayes' Rule

- $P(X|Y)$ = Fraction of the worlds in which X is true given that Y is also true.

$$P(Y, X) = P(Y|X) P(X) \Rightarrow \underbrace{P(Y|X)}_{P(X)} = \frac{P(Y, X)}{P(X)} = \frac{P(X|Y) P(Y)}{P(X)}$$

- For example:

- H = "Having a headache"
- F = "Coming down with flu"
- $P(\text{Headache}|\text{Flu})$ = fraction of flu-inflicted worlds in which you have a headache. How to calculate?

$$P(X) = \sum_y P(X, Y=y) = \sum_y P(X|Y=y) P(Y=y)$$

$$\underbrace{P(A|B, C)}_{P(B, C)} = \frac{P(A, B, C)}{P(B, C)} = \frac{P(B|A, C) \cancel{P(A, C)}}{\cancel{P(B|C)} \cancel{P(C)}}$$

- Definition:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{P(Y|X) P(X)}{P(Y)} = \frac{P(B|A, C) \cancel{P(A, C)}}{\cancel{P(B|C)} \cancel{P(C)}}$$

Corollary:

$$P(X, Y) = P(Y|X) P(X)$$

This is called **Bayes Rule**

Bayes' Rule

- $$P(\text{Headache}|\text{Flu}) = \frac{P(\text{Headache}, \text{Flu})}{P(\text{Flu})}$$
$$= \frac{P(\text{Flu}|\text{Headache})P(\text{Headache})}{P(\text{Flu})}$$

Other cases:

- $$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X|Y)P(Y) + P(X|\neg Y)P(\neg Y)}$$
- $$P(Y = y_i|X) = \frac{P(X|Y)P(Y)}{\sum_{i \in S} P(X|Y = y_i)P(Y=y_i)}$$
- $$P(Y|X, Z) = \frac{P(X|Y, Z)P(Y, Z)}{P(X, Z)} =$$
$$\frac{P(X|Y, Z)P(Y, Z)}{P(X|Y, Z)P(Y, Z) + P(X|\neg Y, Z)P(\neg Y, Z)}$$

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$E[\cdot]$ $\rightarrow$ expected value – weighted avg

Mean and Variance

$E(\cdot)$

- Expectation: The mean value, center of mass, first moment:

$$E_X[g(X)] = \int_{-\infty}^{\infty} \underbrace{g(x)} \underbrace{p_X(x)} dx = \mu$$

$$E[g(x)] = \sum p(x) g(x)$$

- N-th moment: $g(x) = x^n$

- N-th central moment: $g(x) = (x - \mu)^n$

- Mean: $E_X[X] = \int_{-\infty}^{\infty} x p_X(x) dx$

- $E[\alpha X] = \alpha E[X]$

- $E[\alpha + X] = \alpha + E[X]$

$$\text{Var}(X) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = E[(X - E[X])^2]$$

- Variance(Second central moment): $\text{Var}(x) =$

$$E_X[(X - E_X[X])^2] = E_X[X^2] - E_X[X]^2$$

- $\text{Var}(\alpha X) = \alpha^2 \text{Var}(X)$

- $\text{Var}(\alpha + X) = \text{Var}(X)$

Mean and average

$$g(x) = x$$

$$X = [1, 2, 3]$$

$$P(x) = \left[\frac{1}{6}, \frac{2}{6}, \frac{3}{6} \right]$$

$$E[g(x)] = \sum g(x) P(x) =$$

$$= g(x=1) P(x=1) + g(x=2) P(x=2) + g(x=3) P(x=3)$$

$$= 1 * \frac{1}{6} + 2 * \frac{2}{6} + 3 * \frac{3}{6}$$

$$E[g(x)] = \frac{14}{6}$$

$$\begin{aligned} \text{avg}(g(x)) &= \frac{1+2+3}{3} \\ &= 2 \end{aligned}$$

$$E[\cdot] = \text{avg} = \mu = \frac{1}{N} \sum (\cdot)$$

$$X = [1, 2, 2, 3, 3, 3]$$

$g(x)$

$$\text{avg} = \frac{1+2+2+3+3+3}{6} = \frac{14}{6}$$

Variance and average:

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \begin{matrix} \text{height} = h \\ n = 3 \\ d = 1 \end{matrix}$$

$$\mu_h = \frac{1+2+3}{3} = 2$$

$$\text{Var}(X_h) = \frac{1}{N} \sum (x_i - \mu_h)^2$$

For loop

$$\left(\frac{1}{N} \sum (\cdot) \right) = \mu = E[\cdot] \rightarrow E[(x - \mu_h)^2]$$

$$E[(x - E[X_h])^2]$$

$$X \rightarrow \bar{X} \quad \bar{X} = \begin{bmatrix} \bar{h} = h - \mu_h \\ 1 - \mu_h \\ 2 - \mu_h \\ 3 - \mu_h \end{bmatrix}_{3 \times 1} = \begin{bmatrix} \bar{h} \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$\mu_h \bar{h} = 0$

$$\text{Var}(X_h) = \frac{1}{N} \bar{X}^T \bar{X}$$

$1 \times 3 \quad 3 \times 1$

$$\text{Var}(X_h) = \frac{1}{N} [1 - \mu_h \quad 2 - \mu_h \quad 3 - \mu_h] \begin{bmatrix} 1 - \mu_h \\ 2 - \mu_h \\ 3 - \mu_h \end{bmatrix} = \frac{1}{N} [(1 - \mu_h)^2 + (2 - \mu_h)^2 + (3 - \mu_h)^2]$$

Covariance:

$$X = \begin{matrix} & \begin{matrix} h & \text{weight} = w \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \end{matrix}_{n \times d}$$

$$X \rightarrow \bar{X}$$

$$X^{\{1\}} = [1, 4] \quad X_1 = [1, 2, 3]$$

$$\bar{X} =$$

$$\begin{bmatrix} \bar{h} = h - \mu_h & \bar{w} = w - \mu_w \\ 1 - \mu_h & 4 - \mu_w \\ 2 - \mu_h & 5 - \mu_w \\ 3 - \mu_h & 6 - \mu_w \end{bmatrix}$$

$$\mu_{\bar{h}} = 0$$

$$\mu_{\bar{w}} = 0$$

$$\{3\} \rightarrow \text{data point}$$

$$X_2 = 6$$

$$\{2\} \rightarrow \text{feature}$$

$$\text{COV} = \frac{1}{N} \bar{X}_{d \times n}^T \bar{X}_{n \times d} = \frac{1}{w} \begin{bmatrix} \sigma_h^2 & \sigma_{hw} \\ \sigma_{wh} & \sigma_w^2 \end{bmatrix}_{d \times d}$$

$$\sigma_{hw} = \sigma_{wh} \rightarrow \text{Symmetric matrix}$$

$$= \frac{1}{N} \begin{bmatrix} 1 - \mu_h & 2 - \mu_h & 3 - \mu_h \\ 4 - \mu_w & 5 - \mu_w & 6 - \mu_w \end{bmatrix} \begin{bmatrix} 1 - \mu_h \\ 2 - \mu_h \\ 3 - \mu_h \end{bmatrix} \begin{bmatrix} 4 - \mu_w \\ 5 - \mu_w \\ 6 - \mu_w \end{bmatrix} \Rightarrow$$

$$\textcircled{1} \frac{1}{N} [(1 - \mu_h)^2 + (2 - \mu_h)^2 + (3 - \mu_h)^2]$$

Correlation: EDA

Standardization

$$\bar{X} = \begin{bmatrix} \bar{h} & \bar{w} \\ 1-\mu_h & 4-\mu_w \\ 2-\mu_h & 5-\mu_w \\ 3-\mu_h & 6-\mu_w \end{bmatrix} \rightarrow \bar{X}^* = \begin{bmatrix} \bar{h}^* & \bar{w}^* \\ \frac{1-\mu_h}{\sigma_h} & \frac{4-\mu_w}{\sigma_w} \\ \frac{2-\mu_h}{\sigma_h} & \frac{5-\mu_w}{\sigma_w} \\ \frac{3-\mu_h}{\sigma_h} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix}$$

$$\bar{w}^* = \frac{\bar{w}}{\sigma_w}$$

$$Cor = \frac{1}{N} \bar{X}_{d \times n}^{*T} \bar{X}_{n \times d}^* = \begin{bmatrix} h & w \\ h & 1 \leq \sigma_{hw} \leq 1 \\ w & 1 \leq \sigma_{wh} \leq 1 & 1 \end{bmatrix}_{d \times d}$$

$$Cor = \frac{1}{N} \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & \frac{2-\mu_h}{\sigma_h} & \frac{3-\mu_h}{\sigma_h} \\ \frac{4-\mu_w}{\sigma_w} & \frac{5-\mu_w}{\sigma_w} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix} \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & 1 & 1 \\ \frac{2-\mu_h}{\sigma_h} & 1 & 1 \\ \frac{3-\mu_h}{\sigma_h} & 1 & 1 \end{bmatrix}$$

$$\textcircled{1} = \frac{1}{N \sigma_h^2} [(1-\mu_h)^2 + (2-\mu_h)^2 + (3-\mu_h)^2]$$

For Joint Distributions

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \underbrace{E[Z^3]}_0 - \underbrace{E[Z]}_0 E[Z^2] = 0$$

$$\text{Var}(Z) = E[Z^2] - E[Z]^2 \Rightarrow \text{Var}(X) = E[Y] - E[X]^2$$

• Expectation and Covariance: $1 = E[Y] - 0 \Rightarrow$

• $E[X + Y] = E[X] + E[Y]$ $E[a+b+c] = E[a+b] + E[c]$

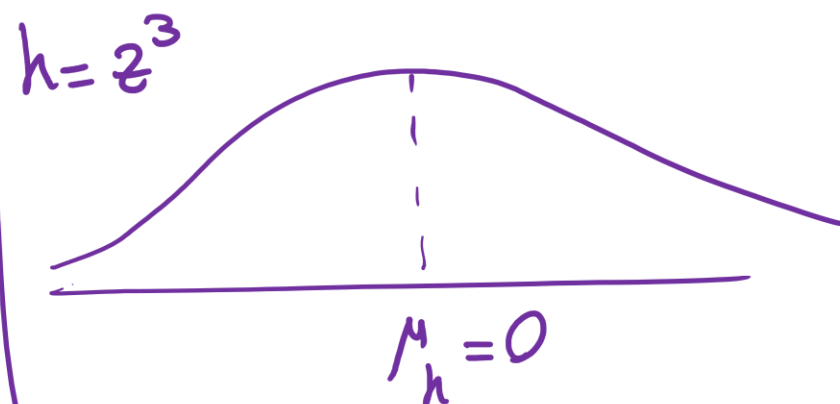
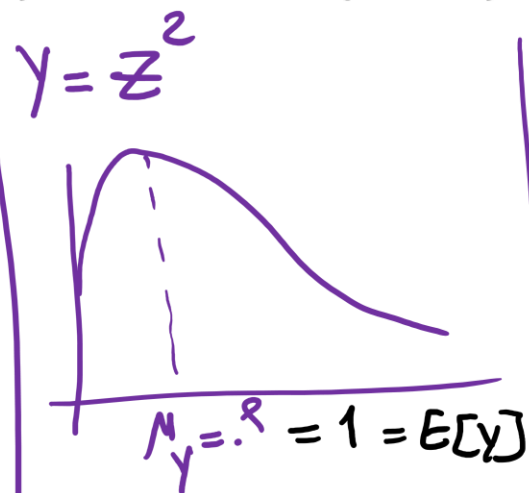
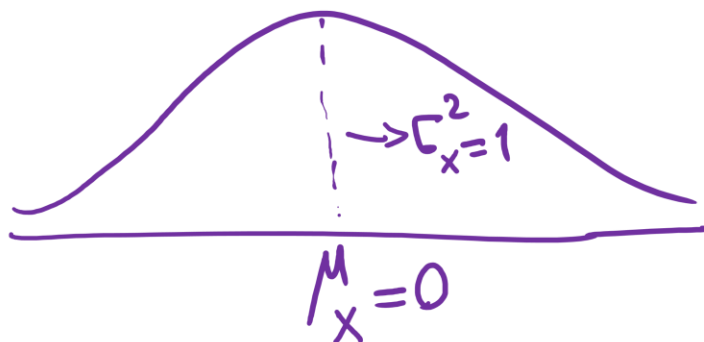
• $\text{cov}(X, Y) = E[(X - E_X[X])(Y - E_Y[Y])] = \underline{E[XY] - E[X]E[Y]}$

• $\text{Var}(X + Y) = \text{Var}(X) + 2\text{cov}(X, Y) + \text{Var}(Y)$

Z is a standard gaussian distribution

$$\mu_Z = 0 \quad \sigma_Z^2 = 1$$

$$X = Z$$



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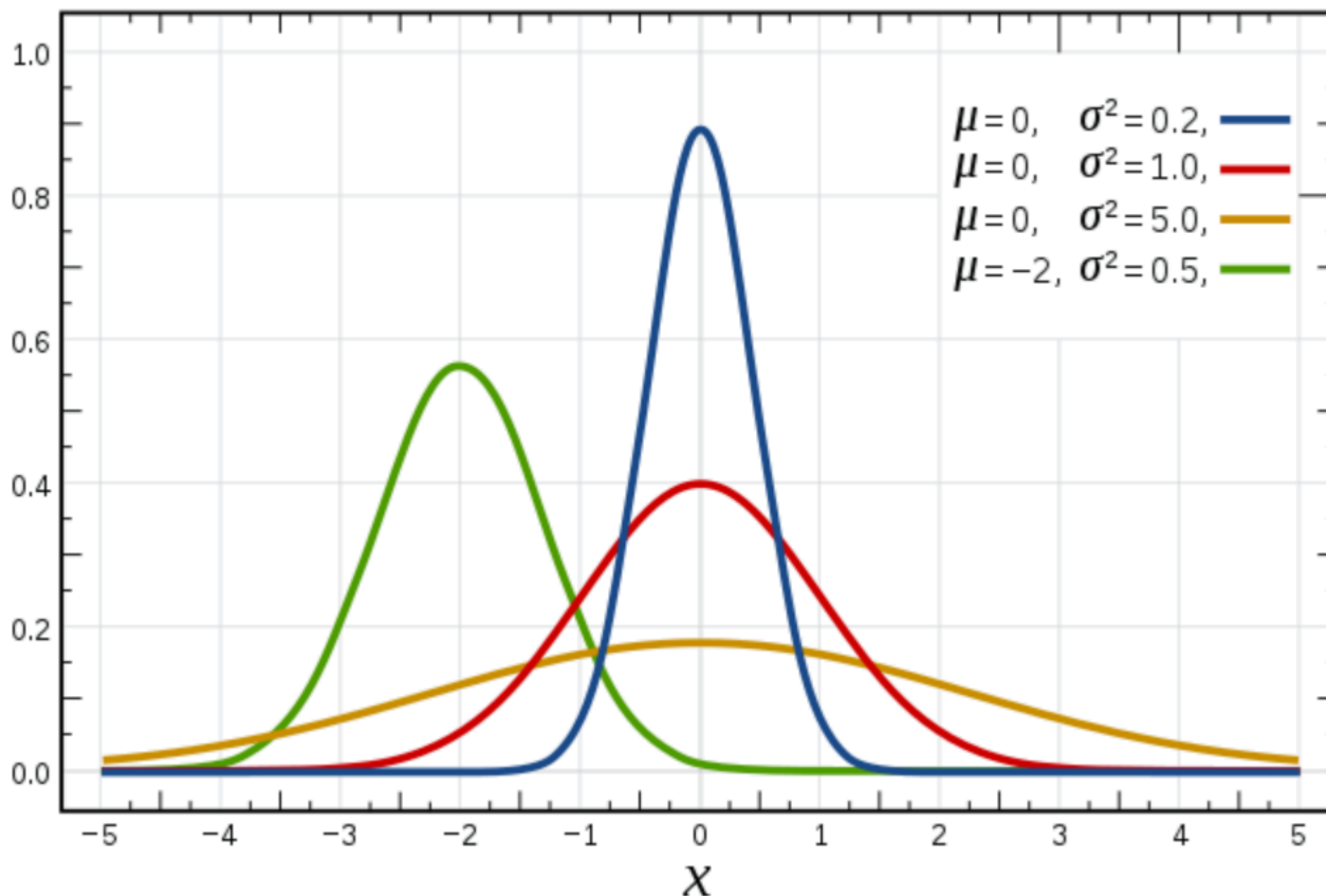
Gaussian Distribution

$\sum^2$ Covariance

$$\{\mu, \sigma^2\} \in \theta$$

univariate

- Gaussian Distribution: $f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- Probability density function



Probability versus likelihood

Prob vs Likelihood

$$P(X=T) = \frac{1}{2}$$

$$T, T, T, H \rightarrow P(X=T) = \frac{3}{4}$$

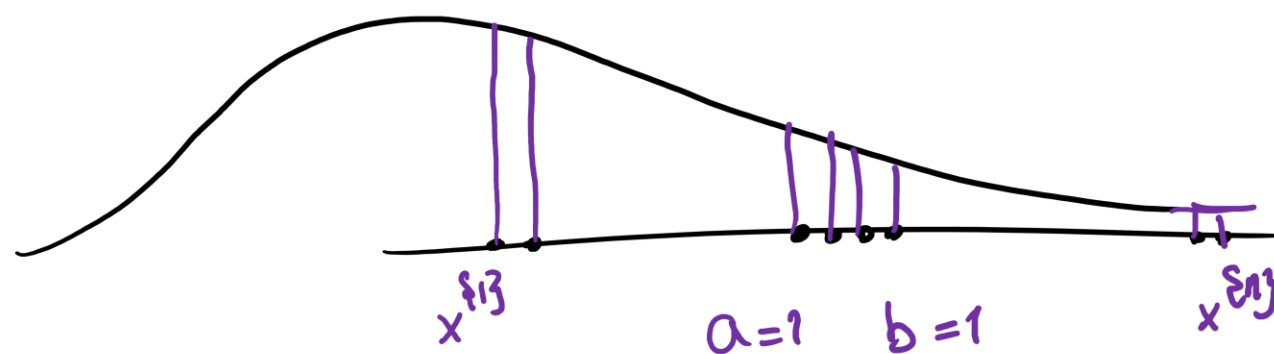
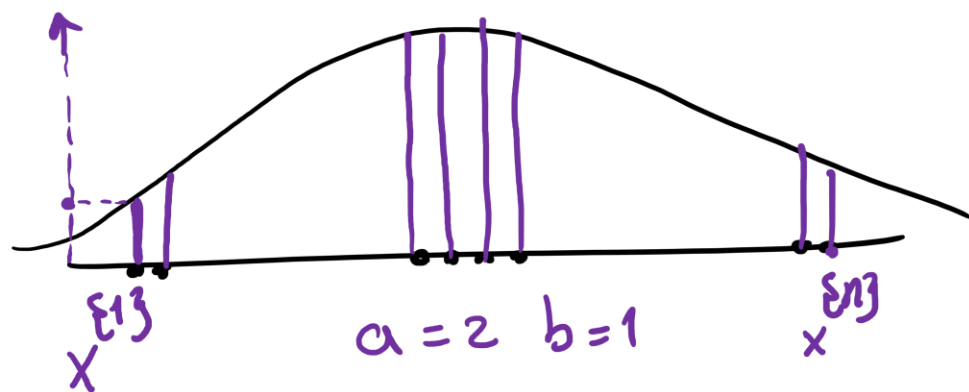
$$P(a,b) = P(a|b) P(b) \\ = P(a) P(b)$$

$$P(a,b,c|\theta) = P(a|\theta) P(b|\theta) P(c|\theta)$$

$$f(x|\theta) = f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad \bigg| \quad f(x|a,b) = \frac{1}{\sqrt{2\pi b^2}} \exp\left[-\frac{(x-a)^2}{2b^2}\right]$$

①

Likelihood or density



$$L(a,b|X) \begin{cases} \rightarrow a=2, b=1 & f(x^{(1)}|a,b) = \dots \\ \rightarrow a=1, b=1 & f(x^{(1)}|a,b) = \dots \end{cases}$$

$$\Rightarrow L(a,b|x^{(1)}, x^{(2)}, \dots, x^{(n)}) = f(x^{(1)}, x^{(2)}, \dots, x^{(n)}|a,b) \\ = f(x^{(1)}|a,b) f(x^{(2)}|a,b) \dots f(x^{(n)}|a,b)$$

Objective function

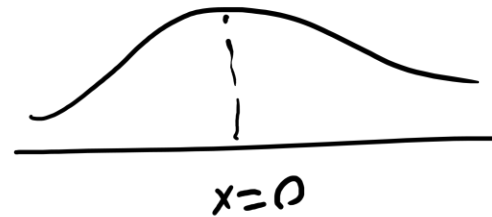
②

Prob vs Likelihood

$$\text{Max } L(\theta|x) = \prod_{i=1}^N f(x^{i3}|\theta) = \prod_{i=1}^N f(x^{i3}|a,b)$$

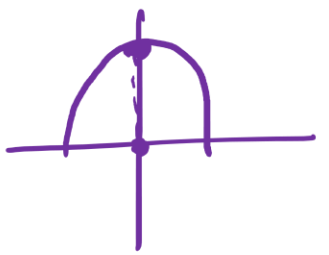
$$a.b \Rightarrow \log(a.b) = \log a + \log b$$

$$\text{Max } \log L(\theta|x) = \ell(\theta|x) = \log f(x^{13}|a,b) + \dots + \log f(x^{N3}|a,b) = \sum_{i=1}^N \log f(x^{i3}|a,b)$$



$$\frac{\partial \ell(a,b|x)}{\partial a} = 0 \Rightarrow a = \frac{1}{N} \sum_{i=1}^N x^{i3} = \mu$$

$$\frac{\partial \ell(a,b|x)}{\partial b} = 0 \Rightarrow b = \sqrt{\frac{1}{N} \sum_{i=1}^N (x^{i3} - \mu)^2}$$

$$f(x) = -x^2 \quad \frac{\partial f(x)}{\partial x} = 0 \Rightarrow 2x=0 \quad x=0$$


Multivariate Gaussian Distribution

$$p(x|\mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left\{-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right\}$$

- Moment Parameterization $\mu = E(X)$

$$\Sigma = \text{Cov}(X) = E[(X - \mu)(X - \mu)^\top]$$

- Mahalanobis Distance $\Delta^2 = (x - \mu)^\top \Sigma^{-1} (x - \mu)$
- Tons of applications (MoG, FA, PPCA, Kalman filter,...)

Properties of Gaussian Distribution

- The **linear transform** of a Gaussian r.v. is a Gaussian. Remember that no matter how x is distributed

$$E(AX + b) = AE(X) + b$$

$$\text{Cov}(AX + b) = A\text{Cov}(X)A^T$$

this means that for Gaussian distributed quantities:

$$X \sim N(\mu, \Sigma) \rightarrow AX + b \sim N(A\mu + b, A\Sigma A^T)$$

- The **sum** of two independent Gaussian r.v. is a Gaussian

$$Y = X_1 + X_2, X_1 \perp X_2 \rightarrow \mu_y = \mu_1 + \mu_2, \Sigma_y = \Sigma_1 + \Sigma_2$$

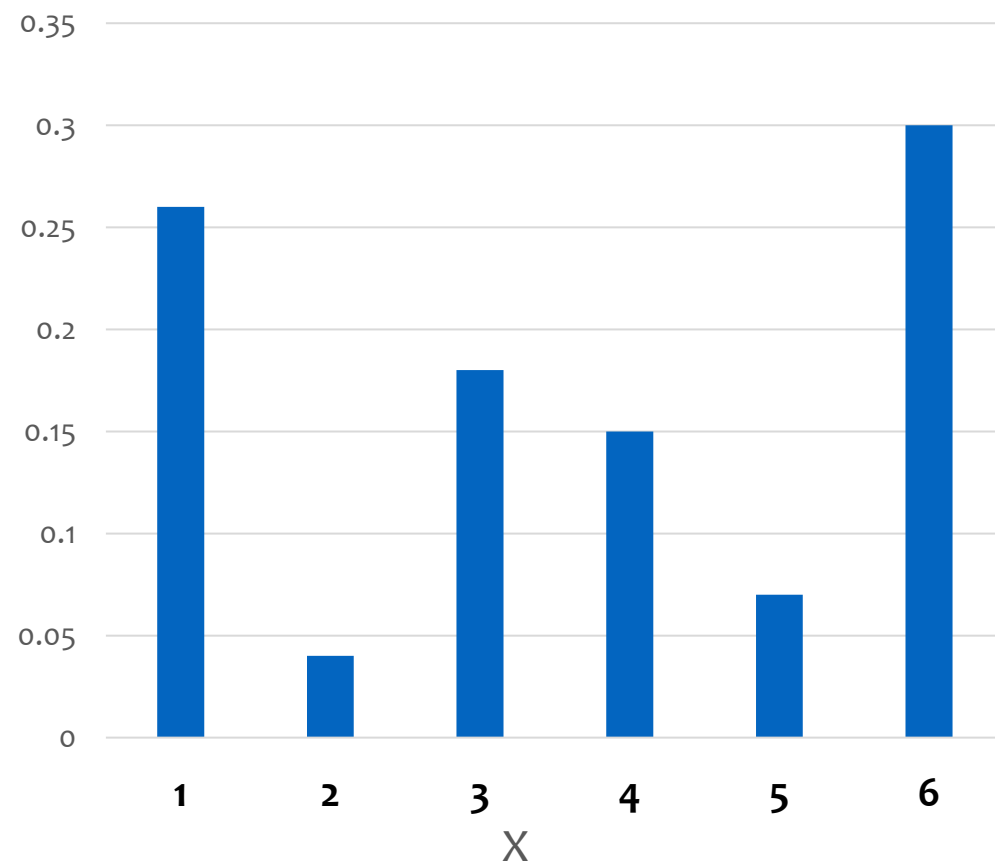
- The **multiplication** of two Gaussian functions is another Gaussian function (although no longer normalized)

$$N(a, A)N(b, B) \propto N(c, C),$$

$$\text{where } C = (A^{-1} + B^{-1})^{-1}, c = CA^{-1}a + CB^{-1}b$$

Central Limit Theorem

Probability mass function of a **biased** dice



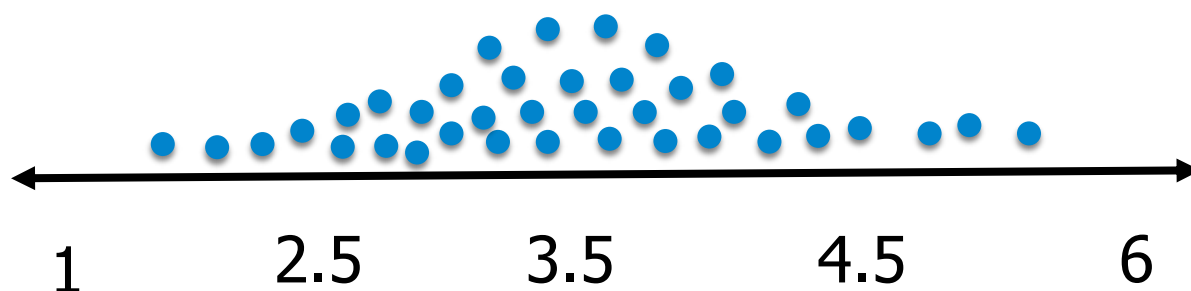
Let's say, I am going to get a sample from this pmf having a size of **$n = 4$**

$$S_1 = \{1,1,1,6\} \Rightarrow E(S_1) = 2.25$$

$$S_2 = \{1,1,3,6\} \Rightarrow E(S_2) = 2.75$$

$\vdots$

$$S_m = \{1,4,6,6\} \Rightarrow E(S_m) = 4.25$$



According to CLT, it will follow a bell curve distribution (normal distribution)

Outline

- Probability Distributions
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation 

Maximum Likelihood Estimation

- Probability: inferring probabilistic quantities for data given fixed models (e.g. prob. of events, marginals, conditionals, etc).
- Statistics: inferring a model given fixed data observations (e.g. clustering, classification, regression).

Main assumption:

Independent and identically distributed random variables
i.i.d

Maximum Likelihood Estimation

For Bernoulli (i.e. flip a coin):

Objective function: $P(x^{\{i\}}|\theta) = \theta^{x^{\{i\}}} (1 - \theta)^{1-x^{\{i\}}}$ $x^{\{i\}} \in \{0,1\}$ or *head, tail*

$$L(\theta|X) = L(\theta|X = x^{\{1\}}, X = x^{\{2\}}, X = x^{\{3\}}, \dots, X = x^{\{n\}})$$

i.i.d assumption

$$L(\theta|X) = \prod_{i=1}^n P(x^{\{i\}}|\theta)$$

$$L(\theta|X) = \prod_{i=1}^n P(x^{\{i\}}|\theta) = \prod_{i=1}^n \theta^{x^{\{i\}}} (1 - \theta)^{1-x^{\{i\}}}$$

$$\begin{aligned} L(\theta|X) &= \theta^{x^{\{1\}}} (1 - \theta)^{1-x^{\{1\}}} \times \theta^{x^{\{2\}}} (1 - \theta)^{1-x^{\{2\}}} \dots \times \theta^{x^{\{n\}}} (1 - \theta)^{1-x^{\{n\}}} = \\ &= \theta^{\sum x^{\{i\}}} (1 - \theta)^{\sum (1-x^{\{i\}})} \end{aligned}$$

We don't like multiplication, let's convert it into summation

What's the trick?

Take the log

$$\log a^b = b \log a$$

$$L(\theta|X) = \theta^{\sum x^{i}} (1 - \theta)^{\sum (1 - x^{i})}$$

$$\frac{\partial (\log \theta)}{\partial \theta} = \frac{1}{\theta}$$

$$\log L(\theta|X) = l(\theta|X) = \log(\theta) \sum_{i=1}^n x^{i} + \log(1 - \theta) \sum_{i=1}^n (1 - x^{i})$$

How to optimize θ ?

$$\frac{\partial l(\theta|X)}{\partial \theta} = 0 \quad \frac{\sum_{i=1}^n x^{i}}{\theta} - \frac{\sum_{i=1}^n (1 - x^{i})}{1 - \theta} = 0$$

$$\theta = \frac{1}{n} \sum_{i=1}^n x^{i}$$